# Angle-Scanned Phase Illumination for Fixed-Focus 3D Fluorescence Imaging with Structural Reliability Analysis

## Supplementary Material

This supplement gives the details of the optical study (Secs. S1 to S4), the reduced-field structural analysis (Secs. S5 to S11), a preliminary single-stack study (Sec. S12) and implementation details (Sec. S13). Section, figure and table numbers of the main paper are written without the prefix S.

## S1. Optical encoders and photon model

**Encoders.** All four encoders share the NA-0.8 objective pupil, the vectorial Richards–Wolf excitation relay, the $0.32\,\mu$m analysis binning, the NA-0.1 emission PSF, the $12 \times 12$ camera, 256 frames and the voxel-integrated assembly; only the field reaching the relay object plane differs. *Phase plate*: tilted plane wave times a fixed random phase plate ($0.32\,\mu$m features, phases uniform on $[0, 2\pi)$) propagated over $D = 6.6\,\mu$m. *No plate*: the same angle scan and propagation without the plate. *In-focus DMD*: 256 random binary amplitude patterns ($0.32\,\mu$m pixels, 50% on) imaged in focus. *Amplitude mask*: the angle scan through a fixed random binary amplitude mask ($0.32\,\mu$m, 50% open) at the plate's position.

**Illuminated aperture.** If the simulated illuminated aperture equals the $30.7\,\mu$m field, its hard edge, $6.6\,\mu$m before the image plane, is itself a defocused amplitude encoder: tilted-beam edges sweep laterally by $z \tan\theta$ and their diffraction ripples reach the field centre, so even the no-plate control acquires depth structure. The comparison of the main paper therefore simulates every encoder on an optical grid twice the field width ($61.4\,\mu$m aperture) and crops the excitation to the field, as in a microscope whose illuminated area exceeds its field of view. This geometry was adopted after a first comparison with the field-sized aperture revealed the effect; that comparison is reported in Sec. S3.

**Photon model.** Each encoder's excitation is normalised to the same mean over all frames and the analysis volume (equal deposited excitation dose). Counts are $\mathrm{Poisson}(\kappa_{\mathrm{frame}} y_i)$ with $\kappa_{\mathrm{frame}} = \kappa_0\, 256/k$. The exposure $\kappa_0$ of each specimen brings the plate to 1000 photons per pixel per frame at the mean signal at $k = 256$ (15 dB); it is computed from the forward model alone. At equal dose the mean detected signal of every encoder is within 4% of the plate's on every test crop with the overfilled aperture (within 17% with the field-sized aperture). Incident power through the pupil (Tab. S1) sets the laser power required, not the specimen dose: with the field-sized aperture the plate passes 28% and the amplitude mask 21% of the incident light.

| encoder | full field | reduced field | build (s) | incident power |
|---|---|---|---|---|
| phase plate | 6.2e-07 | 1.4e-07 | 327 | – |
| no plate | 6.1e-07 | 1.3e-07 | 43 | – |
| in-focus DMD | 6.0e-07 | 1.3e-07 | 42 | – |
| amplitude mask | 6.3e-07 | 1.3e-07 | 36 | – |

| encoder | full field | reduced field | build (s) | incident power |
|---|---|---|---|---|
| phase plate | 8.0e-07 | 1.4e-07 | 111 | 0.277 |
| no plate | 1.4e-06 | 1.3e-07 | 186 | 0.570 |
| in-focus DMD | 6.3e-07 | 1.3e-07 | 104 | 0.378 |
| amplitude mask | 9.6e-07 | 1.4e-07 | 111 | 0.211 |

Table S1. Full-size operators ($36{,}864^2$) for the overfilled (top) and the field-sized aperture (bottom): largest relative $\ell_2$ difference from the live forward model on three random volumes (full and reduced field; tolerance $10^{-5}$), assembly time on one GPU, and the fraction of incident power that passes the pupil.

| encoder | prior | $C = 1$ | $C = 4$ | $C = 8$ |
|---|---|---|---|---|
| phase plate | DnCNN | 2/0 | 0.5/0 | 0.25/0 |
| | sparsity | 1/4 | 0.5/4 | 0.25/4 |
| no plate | DnCNN | 0.5/0 | 0.125/0 | 0.125/0 |
| | sparsity | 0.5/4 | 0.25/4 | 0.25/4 |
| in-focus DMD | DnCNN | 1/0.0625 | 0.25/0 | 0.125/0 |
| | sparsity | 1/4 | 0.25/4 | 0.5/4 |
| amplitude mask | DnCNN | 2/0 | 0.5/0 | 0.5/0 |
| | sparsity | 1/4 | 0.5/4 | 0.5/4 |

Table S2. Selected PnP parameters ($\rho$ scale / L1 scale) on the initial grid, overfilled aperture. Selections at $C = 4, 8$ on the expanded grid are in Tab. S11.

**Specimens and tuning.** Test: two crops from each of six cells of three animals (DEEP$^2$, public under the MIT licence [4]), selected from the reference volumes alone and frozen before any reconstruction. Validation: two crops of a separate vascular volume (DnCNN prior, TV) and a held-out bead layout (sparsity prior). The penalty scale $\rho$ and the L1 scale are selected per encoder and compression by mean validation PSNR (Tab. S2); Sec. S4 repeats the selection with an expanded grid.

## S2. Mechanism and point-emitter diagnostics

**Depth structure of the excitation.** Table S3 summarises, in the central half-field, how the per-pixel temporal excitation

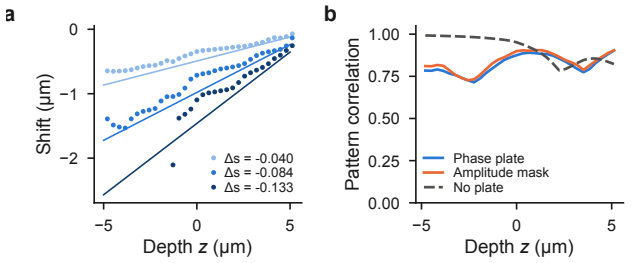



Figure S1. (a) Lateral shift of the plate's excitation between scan pattern 0 and patterns 1–3 (same azimuth) versus depth; lines: thin-screen prediction pivoting at the plate's conjugate plane. (b) Peak normalised correlation between scan patterns 0 and 1 per depth.

signature (its 256 values) decorrelates with depth. Without the plate every depth receives essentially the same sequence. The in-focus DMD decorrelates with depth as fast as the plate but is identical at mirror depths. The plate decorrelates within about $1\,\mu\mathrm{m}$ and its mirror depths are uncorrelated.

**Shear with input tilt.** Figure S1 measures the lateral translation of the plate's excitation between scan angles at every depth. The shift is linear in depth, with slopes 0.74–0.94 of the thin-screen prediction $\Delta s/\sqrt{1-s^2}$ and a pivot 6.2–7.0 µm from focus, against the plate's conjugate plane at $D = 6.6\,\mu\mathrm{m}$. Patterns also partly decorrelate between angles because the pupil passes a different part of the tilted angular spectrum. The amplitude mask's patterns decorrelate between angles like the plate's, whereas the no-plate excitation barely changes.

**Point-emitter diagnostics.** Table S4 uses operator columns for 25 interior lateral positions and 16 depths, with a Gaussian approximation of shot noise at the plate's 15 dB exposure and unknown brightness. These quantities describe the information in the measurements, not optical resolution.

| encoder | contrast (focus) | signature corr. at $\Delta z$ (µm) 0.32 | 0.64 | 1.28 | 2.56 | mirror corr. | border contrast |
|---|---|---|---|---|---|---|---|
| phase plate | 0.71 | 0.86 | 0.53 | 0.09 | 0.05 | 0.05 | 0.71 |
| no plate | 0.18 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 0.25 |
| in-focus DMD | 0.82 | 0.96 | 0.56 | 0.13 | 0.02 | 1.00 | 0.82 |
| defocused amplitude mask | 0.54 | 0.92 | 0.72 | 0.44 | 0.43 | 0.44 | 0.55 |

| encoder | contrast (focus) | signature corr. at $\Delta z$ (µm) 0.32 | 0.64 | 1.28 | 2.56 | mirror corr. | border contrast |
|---|---|---|---|---|---|---|---|
| phase plate | 0.72 | 0.86 | 0.54 | 0.10 | 0.05 | 0.05 | 0.83 |
| no plate | 0.38 | 1.00 | 1.00 | 0.98 | 0.94 | 0.96 | 0.81 |
| in-focus DMD | 0.82 | 0.96 | 0.56 | 0.12 | 0.02 | 1.00 | 0.82 |
| defocused amplitude mask | 0.54 | 0.92 | 0.73 | 0.44 | 0.41 | 0.43 | 0.79 |

Table S3. Excitation statistics in the central half-field for the overfilled (left) and the field-sized aperture (right): lateral contrast at focus, median correlation of per-pixel temporal signatures $\Delta z$ apart and at mirror depths about focus, and the lateral contrast at the field border.

| encoder | $k$ | cos(adjacent) | cos(mirror) | axial CRB (µm) | mirror $d'$ | wrong depth (%) | mirror (%) |
|---|---|---|---|---|---|---|---|
| phase plate | 256 | 0.940 | 0.823 | 0.134 | 10.2 | 1.0 | 0.01 |
| | 32 | 0.943 | 0.828 | 0.137 | 10.2 | 1.6 | 0.04 |
| no plate | 256 | 1.000 | 0.999 | 3.373 | 0.8 | 88.2 | 4.10 |
| | 32 | 1.000 | 0.999 | 3.376 | 0.7 | 88.7 | 4.25 |
| in-focus DMD | 256 | 0.967 | 0.990 | 0.183 | 2.4 | 24.6 | 13.23 |
| | 32 | 0.970 | 0.991 | 0.185 | 2.4 | 25.1 | 13.21 |
| defocused amplitude mask | 256 | 0.966 | 0.904 | 0.171 | 8.0 | 3.1 | 0.06 |
| | 32 | 0.969 | 0.908 | 0.180 | 7.7 | 4.5 | 0.09 |

| encoder | $k$ | cos(adjacent) | cos(mirror) | axial CRB (µm) | mirror $d'$ | wrong depth (%) | mirror (%) |
|---|---|---|---|---|---|---|---|
| phase plate | 256 | 0.941 | 0.807 | 0.113 | 12.4 | 0.9 | 0.01 |
| | 32 | 0.944 | 0.810 | 0.116 | 12.3 | 1.2 | 0.02 |
| no plate | 256 | 0.995 | 0.925 | 0.327 | 9.2 | 34.8 | 0.36 |
| | 32 | 0.996 | 0.915 | 0.361 | 9.1 | 36.8 | 0.37 |
| in-focus DMD | 256 | 0.968 | 0.991 | 0.180 | 2.4 | 24.0 | 12.97 |
| | 32 | 0.970 | 0.991 | 0.183 | 2.4 | 24.5 | 12.91 |
| defocused amplitude mask | 256 | 0.965 | 0.861 | 0.131 | 11.4 | 1.4 | 0.00 |
| | 32 | 0.968 | 0.860 | 0.137 | 11.4 | 2.2 | 0.02 |

Table S4. Point-emitter diagnostics, overfilled (top) and field-sized aperture (bottom): median cosine between operator columns of depth-adjacent ($0.64\,\mu$m) and mirror voxels, axial Cramér–Rao bound, mirror separation $d'$, and Monte Carlo maximum-likelihood depth errors (any wrong depth / the mirror depth).

## S3. Comparison with a field-sized aperture

With the field-sized aperture (Tab. S5 and Fig. S2) the aperture edge gives the no-plate scan and the amplitude mask depth structure: they match or exceed the plate in depth error, while the in-focus DMD result is unchanged. Equalising detected photons instead of deposited dose (Tab. S6) changes the plate and no-plate results by at most 0.1 dB.

| $C$ | encoder | PSNR | FG-SSIM | depth ($\mu$m) | bead ($\mu$m) | ghost |
|---|---|---|---|---|---|---|
| 1 | phase plate | 34.25 | 0.778 | 0.329 | 0.00 | 0.035 |
| | no plate | 34.22 (4/12) | 0.763 (7/12) | 0.254 (1/12) | 0.00 | 0.024 |
| | in-focus DMD | 28.76 (12/12) | 0.485 (12/12) | 2.169 (12/12) | 2.99 | 1.007 |
| | amp. mask | 34.92 (2/12) | 0.805 (1/12) | 0.230 (0/12) | 0.00 | 0.026 |
| 4 | phase plate | 33.23 | 0.712 | 0.424 | 0.00 | 0.042 |
| | no plate | 33.81 (2/12) | 0.744 (3/12) | 0.279 (0/12) | 0.02 | 0.015 |
| | in-focus DMD | 28.70 (11/12) | 0.476 (11/12) | 2.170 (12/12) | 3.29 | 1.010 |
| | amp. mask | 34.39 (0/12) | 0.785 (0/12) | 0.263 (0/12) | 0.00 | 0.038 |
| 8 | phase plate | 32.87 | 0.698 | 0.466 | 0.00 | 0.062 |
| | no plate | 33.03 (4/12) | 0.703 (6/12) | 0.323 (1/12) | 0.04 | 0.019 |
| | in-focus DMD | 28.65 (11/12) | 0.468 (11/12) | 2.170 (12/12) | 3.14 | 1.015 |
| | amp. mask | 33.40 (0/12) | 0.728 (0/12) | 0.317 (0/12) | 0.04 | 0.039 |

Table S5. Encoder comparison with the field-sized aperture (PnP, initial grid). In parentheses: crops on which the plate is better.

| $C$ | encoder | PSNR | FG-SSIM | depth ($\mu$m) | bead ($\mu$m) | ghost |
|---|---|---|---|---|---|---|
| 1 | phase plate | 34.18 | 0.776 | 0.327 | 0.00 | 0.035 |
| | no plate | 34.22 (2/12) | 0.762 (7/12) | 0.253 (1/12) | 0.00 | 0.023 |
| | in-focus DMD | 28.87 (5/5) | 0.493 (5/5) | 2.097 (5/5) | – | – |
| 8 | phase plate | 32.90 | 0.698 | 0.467 | 0.00 | 0.063 |
| | no plate | 33.02 (4/12) | 0.702 (6/12) | 0.322 (1/12) | 0.06 | 0.020 |

Table S6. Equal detected photons instead of equal dose (field-sized aperture, parameters of Tab. S5). The DMD was evaluated on 5 of 12 crops at $C = 1$ and the amplitude mask was not evaluated in this setting.

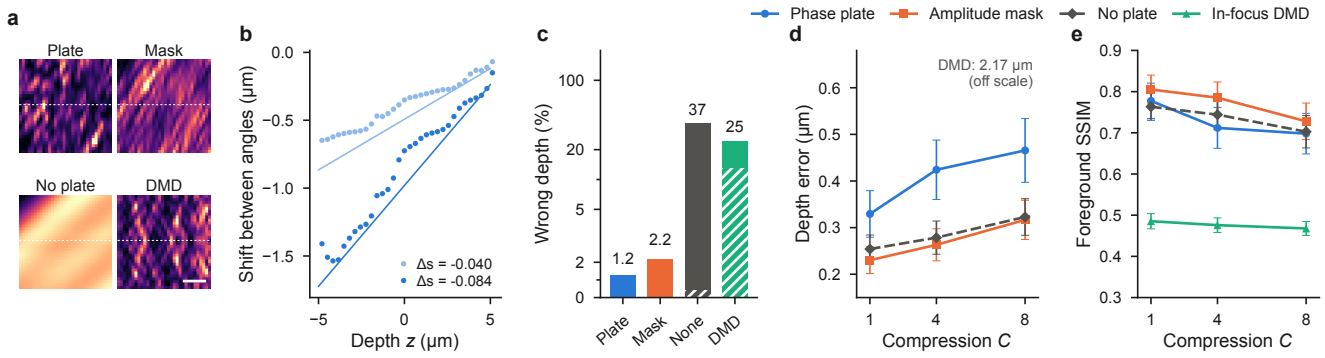



Figure S2. Main-paper Fig. 2 for the field-sized aperture (one realisation, PnP): (a) excitation $xz$ sections, (b) shear with tilt, (c) point emitters assigned a wrong depth at $k = 32$, (d, e) depth error and foreground SSIM versus compression (mean $\pm$ s.e.m. over 12 crops).

## S4. Robustness of the encoder comparison

**Protocol.** Before any result of this study was seen we fixed: three realisations of the phase plate and of the amplitude mask (seeds 0, 101, 202; realisation $r$ draws the central encoder with seed $r$ and the surround of the overfilled aperture with seed $r + 1$, so $r = 0$ is the encoder of the main comparison); a defocus sweep $D \in \{3.3, 6.6, 13.2\}$ $\mu$m for plate realisation 0; compressions $C = 8$ and 4; and an expanded tuning grid for every encoder (DnCNN prior: $\rho$ scale $\in \{2^{-4}, \ldots, 4\} \times$ L1 scale $\in \{0, 0.0625\}$; sparsity prior: $\rho$ scale $\in \{0.125, \ldots, 2\} \times$ L1 scale $\in \{1, 4, 16\}$), extended automatically by halving or doubling $\rho$ (or multiplying the L1 scale by 4) whenever the optimum lay on an edge, at most twice. Every new operator was checked against the live forward model (relative error $\leq 6.3 \times 10^{-7}$). Relay, detector, dose, photon model, noise seeds, direction subsets, objects and metrics are those of the main comparison. Paired differences are computed per crop (mean over five direction subsets), with cell-level bootstrap intervals (6 cells, $10^4$ resamples); realisations are summarised separately and never

pooled as specimens.

**Defocus.** Halving $D$ degrades every metric at $C = 8$ (PSNR worse on 12 of 12 crops) and less at $C = 4$; doubling it leaves PSNR unchanged within the interval and lowers depth error slightly ($0.016$–$0.018\,\mu$m, 9 of 12 crops; Tab. S10). Single-pattern statistics barely change with $D$ (lateral contrast 0.71, depth correlation 0.53 at $0.64\,\mu$m, mirror correlation 0.05), whereas sequence-level quantities do: the median cosine between the $k = 32$ sequences of adjacent depths falls from 0.948 to 0.943, and between mirror depths from 0.854 to 0.823, as $D$ grows, and the shear pivot follows $D$. At $D = 13.2\,\mu$m only the smallest scan step gives a reliable fit; for larger steps the pattern decorrelates within the correlation window.

**Tuning.** No final selection lies on an edge of the extended grid (Tab. S11). For neurons only the no-plate scan at $C = 8$ changed: its test PSNR rose by 0.43 dB and its depth error fell by $0.085\,\mu$m. The sparsity prior for beads moved from L1 scale 4 to 16 for most encoders, which changes bead depth errors by at most $0.02\,\mu$m for the defocused encoders and $0.17\,\mu$m for the DMD (whose beads lie at mirror depths in both cases).

**Total-variation baseline.** We solve $\arg\min_{0\le\mathbf{x}\le1} \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda\,\mathrm{TV}(\mathbf{x})$, isotropic 3D TV of forward differences, by ADMM [2] (penalties 1, conjugate-gradient $x$-update with at most 20 iterations at tolerance $10^{-5}$, 300 iterations) on the same operators and noisy measurements. $\lambda \in \{0.01, 0.03, 0.1, 0.3, 1, 3\}$ is tuned per encoder and compression on the vascular validation crops with the same edge-extension rule. The relative change of $\mathbf{x}$ over the last 10 iterations has medians of $1.3 \times 10^{-3}$–$1.9 \times 10^{-2}$ per encoder and compression (largest at $C = 1$), so most solves do not meet a $10^{-3}$ criterion and we do not claim exact minimisers. Re-solving 12 prespecified test problems at $C = 8$ with 1000 iterations changed PSNR by $-0.03$ dB on average (at most 0.18 dB), foreground SSIM by at most 0.002 and depth error by at most $0.001\,\mu$m; four problems at $C = 1$ changed by at most 0.03 dB and $0.006\,\mu$m.

**Qualitative comparison.** Figures S4 and S5 show the median test crop and the bead layout at $C = 8$ for all encoders. Without the plate, depth is largely lost; the DMD places structure at mirror depths.

| $C$ | encoder | seed | PSNR | FG-SSIM | depth ($\mu$m) | bead ($\mu$m) | ghost |
|---|---|---|---|---|---|---|---|
| 8 | phase plate | 0 | 34.81 | 0.784 | 0.431 | 0.00 | 0.051 |
| 8 | phase plate | 101 | 34.93 | 0.787 | 0.422 | 0.02 | 0.037 |
| 8 | phase plate | 202 | 34.85 | 0.782 | 0.438 | 0.02 | 0.029 |
| 8 | amp. mask | 0 | 33.70 | 0.703 | 0.523 | 0.06 | 0.044 |
| 8 | amp. mask | 101 | 34.37 | 0.766 | 0.419 | 0.17 | 0.067 |
| 8 | amp. mask | 202 | 34.36 | 0.763 | 0.396 | 0.00 | 0.035 |
| 4 | phase plate | 0 | 35.36 | 0.801 | 0.384 | 0.02 | 0.014 |
| 4 | phase plate | 101 | 35.38 | 0.801 | 0.391 | 0.00 | 0.015 |
| 4 | phase plate | 202 | 35.45 | 0.800 | 0.392 | 0.00 | 0.012 |
| 4 | amp. mask | 0 | 34.83 | 0.773 | 0.392 | 0.00 | 0.020 |
| 4 | amp. mask | 101 | 34.95 | 0.782 | 0.382 | 0.02 | 0.019 |
| 4 | amp. mask | 202 | 34.87 | 0.779 | 0.370 | 0.00 | 0.014 |

Table S7. Per-realisation results (multi-slice PnP, 12 test crops; beads: depth error of detected beads and mirror-ghost ratio).

| $C$ | seed | $\Delta$PSNR (dB) | $\Delta$FG-SSIM | $\Delta$depth ($\mu$m) | cells (PSNR) |
|---|---|---|---|---|---|
| 8 | 0 | +1.10 [+0.71, +1.53] (12/12) | +0.081 [+0.062, +0.108] (12/12) | +0.091 [+0.067, +0.122] (11/12) | 6/6 |
| 8 | 101 | +0.56 [+0.28, +0.99] (11/12) | +0.021 [+0.009, +0.038] (12/12) | -0.003 [-0.062, +0.063] (4/12) | 6/6 |
| 8 | 202 | +0.48 [+0.23, +0.78] (10/12) | +0.019 [+0.008, +0.030] (11/12) | -0.042 [-0.096, +0.017] (4/12) | 5/6 |
| 4 | 0 | +0.53 [+0.25, +0.81] (11/12) | +0.028 [+0.019, +0.040] (12/12) | +0.008 [-0.018, +0.034] (7/12) | 6/6 |
| 4 | 101 | +0.43 [+0.19, +0.73] (12/12) | +0.018 [+0.011, +0.031] (12/12) | -0.009 [-0.051, +0.033] (5/12) | 6/6 |
| 4 | 202 | +0.58 [+0.30, +0.91] (12/12) | +0.021 [+0.014, +0.031] (12/12) | -0.022 [-0.059, +0.019] (4/12) | 6/6 |

Table S8. Phase plate minus amplitude mask for matched realisations (positive: plate better; depth: reduction of depth error). Brackets: cell-bootstrap 95% interval; parentheses: crops favouring the plate; last column: cells favouring the plate in PSNR. Over all nine plate/mask pairings, including mismatched seeds, the plate is better in PSNR by 0.44–1.23 dB at $C = 8$ and 0.40–0.62 dB at $C = 4$ and in foreground SSIM by 0.016–0.084, while the depth-error difference ranges from $-0.042$ to $+0.101$ $\mu$m.

| $C$ | seed | SOM004 c1 | SOM004 c2 | SOM005 c1 | SOM006 c1 | SOM006 c2 | SOM006 c3 |
|---|---|---|---|---|---|---|---|
| 8 | 0 | +0.67 | +1.96 | +0.52 | +0.81 | +1.57 | +1.09 |
| 8 | 101 | +0.42 | +0.39 | +0.18 | +0.23 | +1.57 | +0.59 |
| 8 | 202 | +0.38 | +0.45 | +0.36 | -0.02 | +1.12 | +0.61 |
| 4 | 0 | +0.30 | +1.07 | +0.04 | +0.43 | +0.92 | +0.41 |
| 4 | 101 | +0.07 | +0.54 | +0.16 | +0.31 | +1.13 | +0.35 |
| 4 | 202 | +0.21 | +0.79 | +0.31 | +0.15 | +1.23 | +0.79 |

Table S9. Phase plate minus amplitude mask PSNR (dB) per cell (mean over its two crops). 35 of 36 entries are positive. By animal, the plate's PnP PSNR at $C = 8$ spans 34.2–35.9 dB over realisations, with the same encoder ranking in every animal.

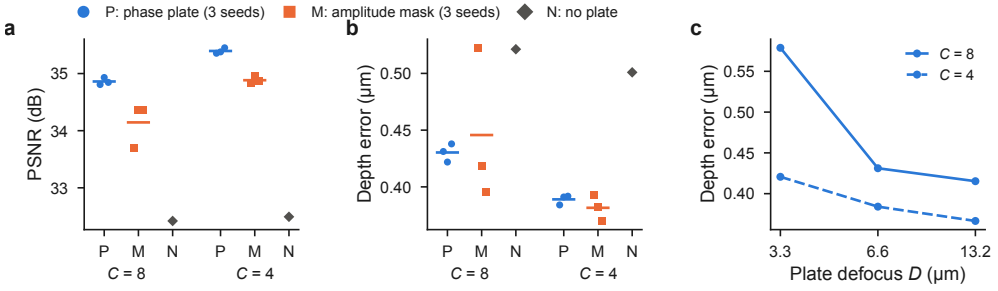



Figure S3. (a, b) PSNR and depth error of the three plate (P) and three amplitude-mask (M) realisations (markers; bar: mean) and of the no-plate scan (N). (c) Plate depth error versus defocus $D$.

| $C$ | $D$ ($\mu$m) | PSNR | FG-SSIM | depth | bead | ΔPSNR vs. 6.6 | wrong depth (%) | adj. cos. | mirror cos. | pivot ($\mu$m) / slope |
|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 3.3 | 33.80 | 0.709 | 0.579 | 0.06 | -1.01 [-1.40, -0.62] | 2.0 | 0.948 | 0.854 | 3.2 / 0.78 |
| 8 | 6.6 | 34.81 | 0.784 | 0.431 | 0.00 | – | 1.6 | 0.943 | 0.827 | 6.5 / 0.81 |
| 8 | 13.2 | 34.69 | 0.783 | 0.415 | 0.09 | -0.11 [-0.30, +0.08] | 1.5 | 0.943 | 0.823 | 15.3 / 0.66 |
| 4 | 3.3 | 34.97 | 0.785 | 0.421 | 0.02 | -0.39 [-0.73, -0.09] | 2.0 | 0.948 | 0.854 | 3.2 / 0.78 |
| 4 | 6.6 | 35.36 | 0.801 | 0.384 | 0.02 | – | 1.6 | 0.943 | 0.827 | 6.5 / 0.81 |
| 4 | 13.2 | 35.26 | 0.802 | 0.367 | 0.00 | -0.09 [-0.23, +0.06] | 1.5 | 0.943 | 0.823 | 15.3 / 0.66 |

Table S10. Plate defocus sweep (realisation 0, PnP). ΔPSNR: paired against $D = 6.6\,\mu$m with cell-bootstrap interval. Wrong depth, adjacent and mirror cosine: point-emitter diagnostics at $k = 32$. Pivot / slope ratio: shear fit between scan patterns 0 and 1 (thin-screen prediction: pivot $D$, ratio 1).

| | | neurons (DnCNN) | | beads (L1) | | neuron test | |
|---|---|---|---|---|---|---|---|
| encoder | $C$ | original | expanded (ext.) | original | expanded (ext.) | PSNR | depth ($\mu$m) |
| phase plate | 8 | 0.25/0 | 0.25/0 (0) | 0.25/4 | 0.5/16 (1) | 34.81 → 34.81 | 0.431 → 0.431 |
| phase plate | 4 | 0.5/0 | 0.5/0 (0) | 0.5/4 | 1/16 (1) | 35.36 → 35.36 | 0.384 → 0.384 |
| no plate | 8 | 0.125/0 | 0.0625/0 (1) | 0.25/4 | 0.25/4 (0) | 31.99 → 32.42 | 0.606 → 0.521 |
| no plate | 4 | 0.125/0 | 0.125/0 (0) | 0.25/4 | 0.125/4 (1) | 32.49 → 32.49 | 0.500 → 0.501 |
| in-focus DMD | 8 | 0.125/0 | 0.125/0 (0) | 0.5/4 | 0.5/4 (0) | 28.83 → 28.83 | 2.169 → 2.169 |
| in-focus DMD | 4 | 0.25/0 | 0.25/0 (0) | 0.25/4 | 1/16 (1) | 28.91 → 28.91 | 2.168 → 2.168 |
| amp. mask | 8 | 0.5/0 | 0.5/0 (0) | 0.5/4 | 0.5/16 (1) | 33.70 → 33.70 | 0.523 → 0.523 |
| amp. mask | 4 | 0.5/0 | 0.5/0 (0) | 0.5/4 | 1/16 (1) | 34.83 → 34.83 | 0.393 → 0.392 |

Table S11. Initial versus expanded tuning ($\rho$ scale / L1 scale; in parentheses the number of automatic grid extensions) and the resulting neuron test means (initial → expanded).

| $C$ | encoder | $\lambda$ | PSNR | FG-SSIM | depth ($\mu$m) | plate adv. PSNR | s | $\Delta x$ |
|---|---|---|---|---|---|---|---|---|
| 8 | phase plate | 0.1 | 37.00 | 0.847 | 0.243 | – | 9.5 | 1.3e-03 |
| 8 | amp. mask | 0.1 | 36.74 | 0.827 | 0.237 | +0.26 [-0.04, +0.55] (9/12) | 11.1 | 1.7e-03 |
| 8 | no plate | 0.03 | 34.34 | 0.676 | 0.342 | +2.66 [+1.54, +3.78] (12/12) | 9.9 | 2.6e-03 |
| 8 | in-focus DMD | 0.1 | 28.53 | 0.473 | 2.311 | +8.47 [+6.74, +9.89] (12/12) | 12.5 | 2.5e-03 |
| 4 | phase plate | 0.3 | 37.34 | 0.852 | 0.223 | – | 20.6 | 2.8e-03 |
| 4 | amp. mask | 0.3 | 36.79 | 0.823 | 0.239 | +0.55 [+0.29, +0.82] (11/12) | 18.9 | 3.8e-03 |
| 4 | no plate | 0.03 | 34.74 | 0.703 | 0.302 | +2.60 [+1.22, +4.01] (12/12) | 11.3 | 5.0e-03 |
| 4 | in-focus DMD | 0.3 | 28.49 | 0.465 | 2.326 | +8.85 [+6.93, +10.42] (12/12) | 17.5 | 6.4e-03 |
| 1 | phase plate | 1 | 38.15 | 0.881 | 0.187 | – | 55.3 | 9.1e-03 |
| 1 | amp. mask | 1 | 37.59 | 0.857 | 0.191 | +0.56 [+0.22, +0.88] (10/12) | 59.9 | 7.8e-03 |
| 1 | no plate | 0.3 | 34.39 | 0.682 | 0.295 | +3.76 [+2.45, +5.21] (12/12) | 55.3 | 1.8e-02 |
| 1 | in-focus DMD | 3 | 28.48 | 0.435 | 2.285 | +9.67 [+7.82, +11.34] (12/12) | 86.2 | 1.9e-02 |

Table S12. TV baseline (realisation 0; 12 test crops, five direction subsets for $C > 1$). Plate adv.: plate minus encoder PSNR with cell-bootstrap interval and crops favouring the plate; s: median solve time on one GPU; $\Delta x$: median relative change over the last 10 iterations.

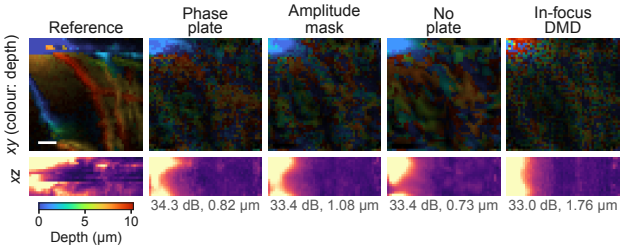



Figure S4. All encoders at $C = 8$ (PnP, one direction subset) on the crop of main-paper Fig. 3: depth-coded lateral projection and $xz$ projection, identical display scale; below: PSNR and depth error of this crop.

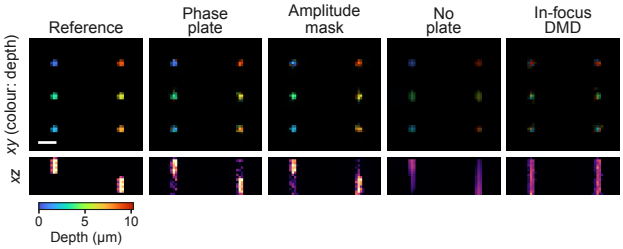



Figure S5. Bead layout at $C = 8$ (PnP, one direction subset) for all encoders, as in Fig. S4.

## S5. Reduced-field structural outcomes per encoder

Table S13 runs the certification pipeline of the main paper on each encoder's reduced operator ($k = 32$, 24 phantoms, seed 0, equal dose). With a field-sized aperture the edge covers most of the $7.68\,\mu\mathrm{m}$ reduced field and gives the no-plate control depth structure; with an overfilled aperture the no-plate control certifies 1 and 6 of 24 queries against 8 and 20 for the plate, and the plate's own counts are unchanged.

| | 15 dB | | 25 dB | |
|---|---|---|---|---|
| encoder | cert. / ambig. | uncertain voxels | cert. / ambig. | uncertain voxels |
| phase plate | 24 / 48 | 7 | 60 / 12 | 0 |
| no plate | 1 / 23 | 54.5 | 6 / 18 | 14 |
| in-focus DMD | 7 / 17 | 17 | 12 / 12 | 3.5 |
| defocused amplitude mask | 8 / 16 | 9 | 20 / 4 | 0 |

| | 15 dB | | 25 dB | |
|---|---|---|---|---|
| encoder | cert. / ambig. | uncertain voxels | cert. / ambig. | uncertain voxels |
| phase plate | 8 / 16 | 9.5 | 20 / 4 | 0 |
| no plate | 7 / 17 | 19 | 18 / 6 | 0 |
| in-focus DMD | 6 / 18 | 19 | 12 / 12 | 4 |
| defocused amplitude mask | 8 / 16 | 14 | 20 / 4 | 0 |

Table S13. Certified / ambiguous queries (of 24) and median uncertain query voxels (of 64) per encoder on the reduced field, overfilled (top) and field-sized aperture (bottom). With the overfilled aperture the plate reproduces the structural counts of the main paper over three seeds exactly.

## S6. Formulation details

**Variables and set.** The reduced field has $N = 576$ voxels; the query region $Q$ is the central $4 \times 4 \times 4$ block (64 voxels, 6-adjacency, 144 edges) and the remaining 512 voxels are free nuisance variables in $[0, 1]$. Each measurement row is rescaled by its largest coefficient before optimisation; all reported residuals are recomputed in unscaled units. For certificates the set is enlarged outward, $\mathcal{C}_\delta = \{\mathbf{x} : \ell_i - \delta\|a_i\|_1 \le a_i^\top \mathbf{x} \le h_i + \delta\|a_i\|_1\}$ with $\delta = 10^{-7}$, so that a certificate on $\mathcal{C}_\delta$ is robust to the solver's feasibility tolerance; witnesses must lie in the unenlarged $\mathcal{C}$ with residual at most $10^{-8}$ (scaled) and $10^{-6}$ (unscaled).

**Minimum bottleneck (separator form).** $b_{\min} = \min_{\mathbf{x}\in\mathcal{C}} \max_P \min_{v\in P} x_v$ is computed through vertex-separator duality. Binary $s_v$ marks the side of $v$ ($s_p = 1$, $s_q = 0$), binary $z_v$ marks membership in the separator, and

$$\begin{aligned} \min\ t\ \text{ s.t. }\ & |s_u - s_v| \le z_u + z_v \quad \forall (u,v) \in E_Q, \\ & x_v \le t + (1 - t_{\mathrm{lo}})(1 - z_v) \quad \forall v \in Q, \qquad \mathbf{x} \in \mathcal{C}. \end{aligned} \tag{S1}$$

Any feasible $(s, z)$ defines a vertex set $S = \{v : z_v = 1\}$ that separates $p$ from $q$, and $t \ge \max_{v\in S} x_v \ge b(\mathbf{x})$; conversely every separator is representable. The query is certified connected when the solver's dual bound on $\mathcal{C}_\delta$ exceeds $\tau + \eta$, $\eta = 10^{-6}$.

**Maximum bottleneck (flow form).** Binary $w_v$ selects a path; a unit flow from $p$ to $q$ is supported on selected vertices ($0 \le f_{uv} \le w_u$, $f_{uv} \le w_v$, flow conservation), and $x_v \ge t - t_{\mathrm{hi}}(1 - w_v)$; maximising $t$ gives $b_{\max}$. The query is certified disconnected when the dual bound is below $\tau - \eta$.

**Decision versions and witnesses.** Fixing $t = \tau \pm \mu$ turns each program into a feasibility problem whose solution, if

any, is a candidate witness $\mathbf{x}_C$ ($b \geq \tau+\mu$) or $\mathbf{x}_D$ ($b \leq \tau-\mu$). Every candidate is re-checked independently: residual in unscaled units, active set at $\tau$, connectivity by breadth-first search and by union–find, and threshold margin $\geq \eta$. Only re-checked pairs count as proven ambiguity.

**Route order and guards.** (i) Per-voxel envelope: $2 \times 64$ LPs on $\mathcal{C}_\delta$, warm-started, with failed solves retried cold and a feasible-point check; (ii) witness-first decision MIPs; (iii) LP path/separator enumeration when at most 16 voxels are uncertain and at most 200 induced paths exist; (iv) the joint MIP verifier. A certificate that contradicts a validated witness or a verified feasible point is reported as numerically undecided, and timeouts are never counted as ambiguity. All programs are solved with HiGHS [3] (single thread, tolerances $10^{-9}$, zero relative MIP gap).

## S7. Reduced operator and phantoms

The reduced operator uses the optics of the full system (532 nm, NA 0.8/0.1, one random plate with $0.32\,\mu\mathrm{m}$ features at $D = 6.6\,\mu\mathrm{m}$, 256 directions, $2 \times 2 \times 2$ patch integration, equal dose) on a smaller field: optical grid $96 \times 96$ at $0.08\,\mu\mathrm{m}$, analysis grid $8 \times 24 \times 24$ at $0.32\,\mu\mathrm{m}$, reconstruction grid $4 \times 12 \times 12$ at $0.64\,\mu\mathrm{m}$, and $3 \times 3$ camera pixels of $2.56\,\mu\mathrm{m}$ per pattern. It agrees with the live simulator to $\leq 3.1 \times 10^{-7}$ relative $\ell_2$ error on 10 volumes, 3 geometries and all 576 re-simulated columns, and $\mathsf{A} = \mathsf{BU}$ holds to $1.3 \times 10^{-7}$; negative controls (column order, axis swap, depth flip) give 8.4%, 7.4% and 3.5% error. All entries are positive; singular values range from 125.3 to 0.058 at $k = 256$ (condition number $2.2 \times 10^3$), and the nested direction subsets have condition numbers $2.0 \times 10^2$, $5.6 \times 10^2$, $8.9 \times 10^5$ and $2.9 \times 10^3$ at $k = 16, 32, 64, 128$. The cosine between operator columns of depth-adjacent voxels is 0.958, 0.869 and 0.802 at separations of 1, 2 and 3 planes. Table S14 lists the 24 phantoms.

| id | family | structure (query region, $4 \times 4 \times 4$) | $p$ | $q$ |
|---|---|---|---|---|
| P01 | connected | straight filament through all depths | (0,5,5) | (3,5,5) |
| P02 | connected | 3D staircase filament, U(0.5,1) | (0,4,4) | (3,7,7) |
| P03 | connected | lateral staircase filament in one plane | (1,4,4) | (1,7,7) |
| P04 | disconnected | two separated filaments (z=0 and z=3) | (0,4,4) | (3,7,7) |
| P05 | disconnected | filament with one missing voxel | (0,5,4) | (3,6,7) |
| P06 | disconnected | filament with one weak (0.3) bridge voxel | (0,6,4) | (3,5,7) |
| P07 | connected | two alternative-depth routes (z=0 and via z=3), both bright | (0,4,5) | (0,7,5) |
| P08 | connected | alternative-depth routes, shallow route dim (0.15-0.25) | (0,4,5) | (0,7,5) |
| P09 | disconnected | laterally crossing filaments at different depths | (0,4,5) | (3,6,7) |
| P10 | connected | crossing filaments joined by a vertical connector | (0,4,5) | (3,6,7) |
| P11 | disconnected | four isolated beads (endpoints on beads) | (0,4,4) | (3,7,7) |
| P12 | connected | bead chain (adjacent beads of varying brightness) | (0,7,4) | (3,7,7) |
| P13 | connected | three alternative routes, intensities 0.5-0.8 | (0,4,4) | (1,4,7) |
| P14 | connected | P01 filament + U(0,0.15) background over the field | (0,5,5) | (3,5,5) |
| P15 | disconnected | P04 + bright distractor blobs outside the ROI | (0,4,4) | (3,7,7) |
| P16 | connected | graded filament $1.0 \rightarrow 0.45$ | (0,7,4) | (3,4,7) |
| P17 | connected | thick (2-voxel) vertical filament | (0,5,5) | (3,6,5) |
| P18 | connected | in-plane filament at z=0 (mirror-depth control: mirror plane empty) | (0,4,6) | (0,7,6) |
| P19 | connected | ring in plane z=1, endpoints opposite | (1,4,4) | (1,7,7) |
| P20 | connected | dim filament (0.42-0.48), just above tau = 0.4 | (0,4,4) | (3,7,7) |
| P21 | connected | random lattice walk (9 voxels) | (0,4,4) | (1,6,5) |
| P22 | unknown | two random walks (connectivity decided by truth BFS) | (0,7,7) | (3,4,4) |
| P23 | unknown | sparse random volume (15% of ROI voxels U(0.3,1)); fixed corners | (0,4,4) | (3,7,7) |
| P24 | unknown | dense random volume U(0,1) over the whole field | (0,4,4) | (3,7,7) |

Table S14. The 24 phantoms. Coordinates are global $(z, a, b)$; the query region is $z \in [0, 3]$, $a, b \in [4, 7]$. For the "unknown" families, connectivity is decided by the truth.

## S8. Reconstruction versus ambiguity: all cases

All 26 reconstructions (two estimators, 13 cases) lie inside $\mathcal{C}(\mathbf{y})$ (Tab. S15 and Fig. S6). In the dense phantom P24 the reconstruction fits the data more closely than the truth (largest normalised residual 0.24 vs. 0.83 at 15 dB, 0.001 vs. 0.83 at 25 dB) while reporting the wrong topology. The total variation of the opposite-topology witnesses is close to that of the truth for the filament phantoms (P01: truth 11.6, witnesses 13.2 and 12.6; P20: 17.5, 17.7 and 20.4; P06: 23.4, 24.3 and 23.0), so these volumes are not rough artefacts. The TV weight of the reconstruction baseline was selected on ten calibration phantoms: $\lambda = 3 \times 10^{-4}$ at 15 dB (mean calibration PSNR 46.9 dB vs. 46.2 dB without TV) and $\lambda = 0$ at 25 dB.




Figure S6. All 13 cases in the layout of main-paper Fig. 4. Certified cases show no opposite-topology volume because none exists in $\mathcal{C}(\mathbf{y})$.

| case | dB | truth | recon. TV | PSNR | recon. LS | consistent C | consistent D | data decide |
|---|---|---|---|---|---|---|---|---|
| P01 | 15 | C 0.60 / 0.61 | C 0.64 / 0.62 | 51.5 | C 0.62 / 0.62 | C 0.49 / 1.00 | D 0.40 / 1.00 | ambiguous |
| P06 | 15 | D 0.30 / 0.72 | D 0.30 / 0.68 | 46.4 | D 0.30 / 0.69 | C 0.40 / 1.00 | D 0.40 / 1.00 | ambiguous |
| P07 | 15 | C 0.72 / 0.83 | C 0.59 / 0.83 | 41.5 | C 0.55 / 0.83 | C 0.41 / 1.00 | D 0.40 / 1.00 | ambiguous |
| P13 | 15 | C 0.54 / 0.79 | C 0.45 / 0.74 | 31.2 | C 0.42 / 0.74 | C 0.40 / 1.00 | D 0.20 / 1.00 | ambiguous |
| P20 | 15 | C 0.42 / 0.54 | C 0.42 / 0.55 | 48.1 | C 0.41 / 0.56 | C 0.45 / 1.00 | D 0.10 / 1.00 | ambiguous |
| P24 | 15 | C 0.43 / 0.83 | D 0.00 / 0.23 | 7.8 | D 0.00 / 0.24 | C 0.40 / 1.00 | D 0.00 / 1.00 | ambiguous |
| P13 | 25 | C 0.54 / 0.79 | C 0.53 / 0.74 | 50.3 | C 0.53 / 0.74 | C 0.56 / 1.00 | D 0.40 / 1.00 | ambiguous |
| P20 | 25 | C 0.42 / 0.54 | C 0.42 / 0.56 | 68.2 | C 0.42 / 0.56 | C 0.41 / 1.00 | D 0.40 / 1.00 | ambiguous |
| P24 | 25 | C 0.43 / 0.83 | D 0.20 / 0.00 | 13.6 | D 0.20 / 0.00 | C 0.40 / 1.00 | D 0.40 / 1.00 | ambiguous |
| P09 | 15 | D 0.00 / 0.61 | D 0.01 / 0.64 | 50.5 | D 0.00 / 0.64 | – | – | cert. D |
| P18 | 15 | C 0.77 / 0.80 | C 0.75 / 0.72 | 58.2 | C 0.76 / 0.70 | – | – | cert. C |
| P06 | 25 | D 0.30 / 0.72 | D 0.30 / 0.69 | 65.3 | D 0.30 / 0.69 | – | – | cert. D |
| P07 | 25 | C 0.72 / 0.83 | C 0.70 / 0.83 | 59.1 | C 0.70 / 0.83 | – | – | cert. C |

Table S15. All 13 reduced-field cases. Each entry: topology at $\tau = 0.4$ (C/D), bottleneck $b$, and the largest normalised residual $\max_i |(\mathbf{A}\mathbf{x} - \mathbf{y})_i|/z\sigma$ (inside $\mathcal{C}$ iff $\leq 1$). Reconstruction PSNR is against the coarse truth.

## S9. Ambiguity versus directions and photon budget

Figure S7 and Tab. S16 report, for the queries that are proven ambiguous with $k = 32$ directions, how many become certified with $k$ directions (row confidence set, $\tau = 0.4$, three seeds pooled, after the 600 s rerun). The decision rule (primary metric: certified fraction at $k = 256$ with fixed total photons; outcomes A $\geq 0.30$, B 0.10–0.30, C $< 0.10$) was fixed before any aggregate was computed; the outcome is C.

**Numerical validation.** 1776 problems were solved with no errors and no wrong certificates. The 110 problems that hit the 60 s limit were rerun at 600 s with nothing else changed: 59 became proven ambiguities, 5 became certificates and 1 numerical boundary case remained. The 1325 stored witness volumes were re-validated on independently rebuilt problems (maximum unscaled residual $9.6 \times 10^{-10}$), and all 18 certificates not issued by the envelope were re-derived by the other exact route. Total compute: 696 CPU-hours.

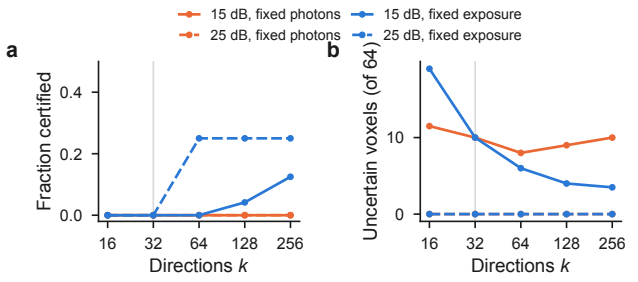



Figure S7. (a) Fraction of the queries proven ambiguous at $k = 32$ that become certified with $k$ directions, with fixed total photons or fixed per-frame exposure ($8\times$ photons at $k = 256$). (b) Median uncertain query voxels over all 72 queries. Three seeds; no query remained undecided.

| $k$ | 16 | 32 | 64 | 128 | 256 |
|---|---|---|---|---|---|
| total dose / baseline | 1× | 1× | 1× | 1× | 1× |
| certified | 0 | 0 | 0 | 0 | 0 |
| proven ambiguous | 48 | 48 | 48 | 48 | 48 |
| undecided | 0 | 0 | 0 | 0 | 0 |
| median uncertain voxels | 11.5 | 10 | 8 | 9 | 10 |
| mean bound width | 0.338 | 0.306 | 0.291 | 0.301 | 0.320 |

(a) 15 dB, fixed total photons ($n = 48$)

| $k$ | 16 | 32 | 64 | 128 | 256 |
|---|---|---|---|---|---|
| total dose / baseline | 1× | 1× | 1× | 1× | 1× |
| certified | 0 | 0 | 0 | 0 | 0 |
| proven ambiguous | 12 | 12 | 12 | 12 | 12 |
| undecided | 0 | 0 | 0 | 0 | 0 |
| median uncertain voxels | 0 | 0 | 0 | 0 | 0 |
| mean bound width | 0.132 | 0.114 | 0.099 | 0.095 | 0.096 |

(b) 25 dB, fixed total photons ($n = 12$)

| $k$ | 16 | 32 | 64 | 128 | 256 |
|---|---|---|---|---|---|
| total dose / baseline | 0.5× | 1× | 2× | 4× | 8× |
| certified | 0 | 0 | 0 | 2 | 6 |
| proven ambiguous | 48 | 48 | 48 | 46 | 42 |
| undecided | 0 | 0 | 0 | 0 | 0 |
| median uncertain voxels | 19 | 10 | 6 | 4 | 3.5 |
| mean bound width | 0.392 | 0.306 | 0.245 | 0.211 | 0.184 |

(c) 15 dB, fixed per-frame exposure ($n = 48$)

| $k$ | 16 | 32 | 64 | 128 | 256 |
|---|---|---|---|---|---|
| total dose / baseline | 0.5× | 1× | 2× | 4× | 8× |
| certified | 0 | 0 | 3 | 3 | 3 |
| proven ambiguous | 12 | 12 | 9 | 9 | 9 |
| undecided | 0 | 0 | 0 | 0 | 0 |
| median uncertain voxels | 0 | 0 | 0 | 0 | 0 |
| mean bound width | 0.149 | 0.114 | 0.087 | 0.073 | 0.065 |

(d) 25 dB, fixed per-frame exposure ($n = 12$)

Table S16. Outcomes per number of directions $k$. "Median uncertain voxels" and "mean bound width" describe the per-voxel envelope over all 72 queries.

| set | dB | photons | ambig. $k$=32 | certified $k$=256 | ambiguous | undecided | truth $\notin \mathcal{C}$ |
|---|---|---|---|---|---|---|---|
| row+SVD | 15 | 8× | 16 | 4 | 11 | 1 | 3 |
| row+SVD | 15 | fixed | 16 | 0 | 15 | 1 | 3 |
| row+SVD | 25 | 8× | 4 | 1 | 3 | 0 | 3 |
| row+SVD | 25 | fixed | 4 | 0 | 4 | 0 | 3 |
| Poisson (Garwood) | 15 | 8× | 17 | 2 | 15 | 0 | 1 |
| Poisson (Garwood) | 15 | fixed | 17 | 0 | 17 | 0 | 1 |
| Poisson (Garwood) | 25 | 8× | 5 | 2 | 3 | 0 | 0 |
| Poisson (Garwood) | 25 | fixed | 5 | 0 | 5 | 0 | 0 |

Table S17. Secondary confidence sets (seed 0, $k \in \{32, 256\}$). Row+SVD adds 576 calibrated projections of the 2304 rows; Poisson uses Garwood intervals at the same family level.

## S10. Total-variation prior

$\mathcal{C}_R(\mathbf{y}) = \{\mathbf{x} \in \mathcal{C}(\mathbf{y}) : \sum_{(u,v)\in E} |x_u - x_v| \le R\}$ over the 1488 6-neighbour edges of the reduced field is encoded exactly with one auxiliary variable $e_{uv} \ge |x_u - x_v|$ per edge and $\sum e_{uv} \le R$. $R_{\mathrm{cal}} = 31.9$ is the largest TV among ten calibration phantoms and was fixed before any test case. The certification pipeline (600 s per solve) was run on the nine ambiguous cases of Sec. S8 at $R_{\mathrm{cal}}$, $2R_{\mathrm{cal}}$ and $4R_{\mathrm{cal}}$ (Tab. S18). In all 18 runs whose budget the truth satisfies, two validated opposite-topology witnesses exist inside $\mathcal{C}_R$, and they are as regular as the truth (P20 at 25 dB: TV 17.3 and 17.5 vs. 17.5; P01 at 15 dB: 12.2 and 12.9 vs. 11.6). No run produced a certificate. Only TV was tested; a neurite-specific learned prior was not.

| case | dB | TV(truth) | $R_{\mathrm{cal}}$ | $2R_{\mathrm{cal}}$ | $4R_{\mathrm{cal}}$ |
|---|---|---|---|---|---|
| P01 | 15 | 11.6 | ambiguous | ambiguous | ambiguous |
| P06 | 15 | 23.4 | ambiguous | ambiguous | ambiguous |
| P07 | 15 | 33.9 | ambiguous† | ambiguous | ambiguous |
| P13 | 15 | 44.3 | empty set† | ambiguous | ambiguous |
| P20 | 15 | 17.5 | ambiguous | ambiguous | ambiguous |
| P24 | 15 | 454.0 | ambiguous† | ambiguous† | ambiguous† |
| P13 | 25 | 44.3 | empty set† | ambiguous | ambiguous |
| P20 | 25 | 17.5 | ambiguous | ambiguous | ambiguous |
| P24 | 25 | 454.0 | empty set† | empty set† | empty set† |

Table S18. Outcomes under a TV budget. †: the true object violates the budget. "Empty set": no volume satisfies both the data and the prior.

## S11. Fine-grid model mismatch

Fine objects live on the $8 \times 24 \times 24$ analysis grid: *sub-voxel occupancy* (each voxel's value carried by 4 of its 8 sub-voxels at twice the value, so $\mathsf{R}\mathbf{x}_f$ equals the coarse truth)

| fine structure | dB | noise only | + mismatch (max) | + mismatch (99.9%) |
|---|---|---|---|---|
| sub-voxel | 15 | 0/24 | 21/24 | 21/24 |
| sub-voxel | 25 | 0/24 | 21/24 | 21/24 |
| multiplicative | 15 | 1/24 | 22/24 | 22/24 |
| multiplicative | 25 | 0/24 | 22/24 | 20/24 |

Table S19. Held-out coverage of the coarse truth (8 phantoms $\times$ 3 seeds). All failures with the allowance come from phantom P18 (a filament confined to the outermost depth plane), whose residual reaches 25%, outside the calibration range.

and a *multiplicative* model $\mathbf{x}_f = \mathrm{clip}(\mathsf{U}\mathbf{x} \cdot U(0,2), 0, 1)$. Measurements are $\mathsf{B}\mathbf{x}_f$ from the fine-grid simulator plus the Gaussian noise of the main study. The allowance $\epsilon_i = c(|y_i| + z\sigma)$ uses $c$ calibrated on ten calibration phantoms $\times$ three seeds: $c = 0.78$ / $0.58$ (sub-voxel; maximum / 99.9% row quantile) and $0.37$ / $0.25$ (multiplicative). Calibration residuals $\|\mathsf{B}\mathbf{x}_f - \mathsf{A}\mathsf{R}\mathbf{x}_f\|/\|\mathsf{B}\mathbf{x}_f\|$ were 5.8–13.0% and 2.3–7.1%.

| fine structure | dB | model | $n$ | truth $\in \mathcal{C}$ | empty | certified | ambiguous | undecided | contradicts truth |
|---|---|---|---|---|---|---|---|---|---|
| multiplicative | 15 | noise | 8 | 0 | 7 | 0 | 1 | 0 | 0 |
| multiplicative | 15 | noise + allowance (max) | 8 | 8 | 0 | 0 | 8 | 0 | 0 |
| multiplicative | 15 | noise + allowance (99.9%) | 8 | 8 | 0 | 0 | 8 | 0 | 0 |
| multiplicative | 25 | noise | 8 | 0 | 7 | 0 | 1 | 0 | 0 |
| multiplicative | 25 | noise + allowance (max) | 8 | 8 | 0 | 0 | 8 | 0 | 0 |
| multiplicative | 25 | noise + allowance (99.9%) | 8 | 7 | 0 | 0 | 8 | 0 | 0 |
| multiplicative | exact | exact | 8 | 0 | 7 | 0 | 1 | 0 | 0 |
| sub-voxel | 15 | noise | 8 | 0 | 7 | 0 | 1 | 0 | 0 |
| sub-voxel | 15 | noise + allowance (max) | 8 | 7 | 0 | 0 | 8 | 0 | 0 |
| sub-voxel | 15 | noise + allowance (99.9%) | 8 | 7 | 0 | 0 | 8 | 0 | 0 |
| sub-voxel | 25 | noise | 8 | 0 | 7 | 0 | 1 | 0 | 0 |
| sub-voxel | 25 | noise + allowance (max) | 8 | 7 | 0 | 0 | 8 | 0 | 0 |
| sub-voxel | 25 | noise + allowance (99.9%) | 8 | 7 | 0 | 1 | 7 | 0 | 1 |
| sub-voxel | exact | exact | 8 | 0 | 8 | 0 | 0 | 0 | 0 |

Table S20. Query outcomes under model mismatch (seed 0). "Contradicts truth" counts certificates opposite to the true connectivity regardless of coverage; the single case (P18, sub-voxel, 25 dB, 99.9% rule) occurred when the allowance did not cover the truth.

## S12. Preliminary single-stack study

Before the encoder comparison, the phase-plate system was evaluated on eight prespecified crops of *one* neuron stack with a Gaussian relative-noise model at 15 dB and the field-sized aperture (Tab. S21); the main paper uses an independent crop set from six cells. On a held-out crop of that study, voxel integration lowers the reconstruction error throughout the iterations relative to a coarse-grid simulation (Fig. S8), and foreground SSIM is 0.913, 0.832 and 0.761 at $C = 1$, 16 and 32 (depth errors $0.177$, $0.354$ and $0.443\,\mu$m); the higher compressions are exploratory and not part of the test-set evaluation.

| $C$ | depth err. ($\mu$m) | FG-SSIM | $\Delta$depth ($\mu$m) | $\Delta$PSNR (dB) |
|---|---|---|---|---|
| 1 | $0.497 \pm 0.183$ | $0.679 \pm 0.238$ | – | – |
| 4 | $0.550 \pm 0.170$ | $0.652 \pm 0.252$ | $+0.053 \pm 0.016$ | $-0.45 \pm 0.35$ |
| 8 | $0.626 \pm 0.207$ | $0.588 \pm 0.273$ | $+0.129 \pm 0.032$ | $-0.91 \pm 0.45$ |

Table S21. Preliminary study: eight crops of one stack, phase plate, mean $\pm$ s.d. over crops of five-subset averages; $\Delta$: paired change against each crop's own $C = 1$ result.

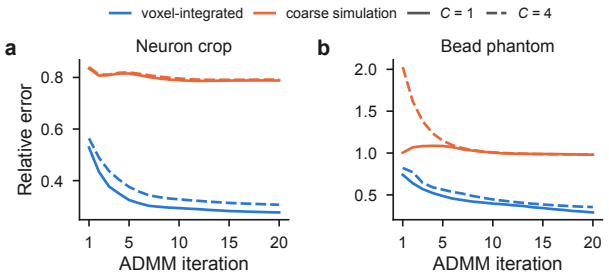



Figure S8. Relative reconstruction error over ADMM iterations with the voxel-integrated and the coarse-simulation operator, on (a) a neuron crop and (b) a bead phantom of the preliminary study (mean of three noise realisations).

## S13. Implementation details

Optical simulation, operator assembly and reconstruction run in PyTorch and JAX on NVIDIA RTX 6000 Ada GPUs; PnP and TV reconstruction use SCICO [1] with a pretrained 17-layer DnCNN [5]. A full-size operator is assembled in 0.6–7 minutes; a PnP reconstruction takes 2–9 s and a TV reconstruction 9–150 s on one GPU. Certification uses HiGHS [3] on CPUs with BLAS threads fixed to one. All tuned quantities of the structural study (TV weight, TV budget, mismatch allowance) were selected on ten calibration phan-

toms (P02, P03, P04, P05, P08, P10, P12, P16, P17, P21)
and fixed before any test case was run.

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